

Chapter 1

Introduction to Spatial Data Analysis

1.1. Introduction to Spatial Data Analysis

- Process of examining, modeling, and interpreting data with a **geographical/spatial component**
- Identifies **patterns, trends, and relationships** not visible in non-spatial data
- Focuses on the importance of **location** (“where”)
- Helps explain **why phenomena occur in specific places**
- Analyzes relationships between features across space
- Supports better decision-making using location-based insights

Applications of Spatial Data Analysis (SDA)

- Widely used in multiple fields:
 - Environmental science
 - Urban planning
 - Public health
 - Business and market analysis
- Key distinction from other data analysis methods:
 - **Location is central** to the analysis
 - Focus on **spatial relationships** between places
 - Examines how nearby or connected locations influence each other

Why Spatial Data Analysis?

- To identify spatial patterns and trends.
- To make decisions based on geographic relationships.
- To perform modeling and prediction using location-based data.

1.2 Terminology Used in Spatial Data Analysis

Understanding the core terminology is crucial. Here are some of the most common terms you'll encounter:

- **Spatial Data:** Information that has a geographic or spatial reference, such as coordinates (latitude and longitude) or addresses. This could be the coordinates of a building, the boundary of a city, or the temperature readings at specific locations.
- **Attribute Data:** The non-spatial information associated with spatial data. For a point representing a school, the attribute data might include the school's name, number of students, and address.

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- **Geographic Information System (GIS):** A system designed to capture, store, manipulate, analyze, manage, and present all types of geographical data. GIS software, like QGIS or ArcGIS, is the primary tool for spatial data analysis.
- **Georeferencing:** The process of aligning data to a known coordinate system, so that it has a real-world location. E.g, Making an old scanned map line up correctly with modern satellite imagery.
- **Coordinate System:** A framework used to define the locations of points in a two- or three-dimensional space. The most common types are :

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- **Geographic Coordinate Systems (GCS)**, which a 3D spherical surface to locate positions on Earth using latitude and longitude, and **Projected Coordinate Systems (PCS)**, A system that projects the Earth's curved surface onto a flat 2D plane for mapping and measurement using linear units (meters, feet, etc.).
- **Projection:** The mathematical process of transferring features from a curved, three-dimensional surface (the Earth) to a flat, two-dimensional surface (a map).
- **Geocoding** is the process of converting a **text-based description of a location**: such as a street **address**, a place name (like a park or landmark), or a postal code into its corresponding **geographic coordinates** (typically latitude and longitude).
- **Spatial Autocorrelation:** The degree to which a set of features and their associated values are similar in location. **Positive spatial autocorrelation** means similar values are clustered together, while **negative spatial autocorrelation** means dissimilar values are clustered.

1.3. Spatial Data Structures and Formats

- Methods used to **organize and store GIS data**
- Define how geographic information is **structured and represented**

Two Primary ways of Structuring & representing Spatial Data are:

- **Vector Model**
- **Raster Model**
- The **Vector Data Model** represents geographic features as discrete, non-continuous elements using geometric shapes: **points**, **lines** (or polylines), and **polygons**. This model is excellent for representing features with distinct boundaries and attributes, such as property lines, roads, or wells.

Vector Structure Components

- **Point:** Represent a specific location, no size or shape. A single pair of (x,y) coordinates (or x,y,z for 3D) representing a specific location with no dimension. Used for features like **telephone poles, well locations, or cities** at small scales.
- **Line (or Polyline):** A sequence of connected vertices (points) representing linear features that have length but no area. Used for features like **roads, rivers, or rail lines**.
- **Polygon:** A sequence of connected vertices where the first and last vertex are the same, forming an enclosed area. Used for features like **lakes, buildings, land parcels, or country boundaries**.
- **Topology:** Defines the spatial relationships between features (e.g., connectivity, adjacency, and containment).

Common Vector Formats

- **Shapefile (.shp):** A very common and widely supported format, though it's actually a collection of several files (e.g., .shp, .dbf, .shx).
- **GeoJSON:** A lightweight, open standard format based on JavaScript Object Notation (JSON), often used for web-based mapping applications.
- **KML (Keyhole Markup Language):** An XML-based format for geographic data, primarily used with applications like Google Earth.

Raster Data Structure and Formats

- The **Raster Data Model** represents geographic space as a regular grid of cells, or **pixels**, organized into rows and columns. Each cell holds a single value representing the characteristic of the area it covers. This model is excellent for representing continuous data and imagery.
- Easy to overlay and perform **mathematical operations**
- Can become **very large** at high resolution. And less precise for **linear features** (roads, rivers) compared to vector data.

Examples of Raster Data

- **Satellite images** – each pixel represents a reflectance value in different bands.
- **Digital Elevation Models (DEM)** – each pixel shows the elevation at that point.
- **Land use/land cover maps** – each pixel is classified (forest, water, urban, etc.).
- **Temperature maps** – pixel values indicate temperature readings.

Raster Structure Components

- **Grid/Matrix:** The fundamental structure where data is stored in cells (pixels).
- **Pixel (Cell):** The smallest, single unit of information in a raster dataset, holding one value. The basic unit of raster data. Each cell has a value representing some attribute (e.g., elevation, temperature, land cover). All cells are square and uniform in size.
- E.g., A 30x30 meter square of land in a satellite image.
- **Cell Value:** The attribute data for a specific area covered by the pixel (e.g., a number representing meters above sea level, a land-use code, or a color value).
- **Resolution:** The size of the area represented by a single raster cell (spatial resolution) or the detail level of the data (e.g., 30m x 30m). Finer resolution means smaller pixels, more detail, and a larger file size. E.g., A 10-meter resolution image provides finer detail than a 30-meter one.
- **Bands:** Rasters can be multi-band, where each pixel has multiple values (layers) stored for the same location, such as the Red, Green, Blue, and Infrared bands in a satellite image.

Common Raster Formats

- **GeoTIFF (.tif):** A standard image format that includes geo-referencing information.
- **ASCII Grid:** A simple text-based format often used for sharing raster data.
- **JPEG2000 (.jp2):** A modern image compression standard that can include georeferencing information.

Importing and Exporting Spatial Data

- Importing and exporting spatial data involves converting data between different formats and coordinate systems so it can be used or shared across various Geographical Information Systems (GIS) software and other platforms. This process is crucial for data compatibility and interoperability.
- **Importing** is the process of loading spatial data from an external file format (like a Shapefile, GeoTIFF, or CSV) into a GIS software's internal structure or database.

Key Steps in the Import Process:

- **Selection of Source File:** Identify the external file containing the spatial data.
- **Format Conversion (if necessary):** The GIS software reads the external format (e.g., GeoJSON) and converts its geometry and attributes into the internal data model (Vector or Raster).
- **Coordinate System Transformation:** This is often the most critical step. If the source data's **Coordinate Reference System (CRS)** is different from the project's CRS, the software must perform a transformation to ensure the data aligns correctly on the map. Data without a defined CRS (unprojected data) will not align correctly and is often referred to as "unreferenced" or "unprojected" data.
- **Data Schema Mapping:** Matching the attribute columns in the source file to the structure required by the target GIS layer or database.

Common Import Scenarios:

- **Importing a CSV/Spreadsheet:** Data with latitude and longitude columns can be imported by telling the GIS software to use those columns as the X and Y coordinates to create a new **point** layer.
- **Importing a CSV/Spreadsheet to R:** Import the `Data(read_csv())` > Convert to Spatial Object(`st_as_sf()`) > Assign proper **coordinate reference system (CRS)**.
- **Importing CAD Data (DWG/DXF):** Converting precise architectural or engineering

Data Export

- **Exporting** is the process of saving data from a GIS layer or database into an external file format for use by others, for archival, or for transfer to another application drawings into GIS vector features.

Key Steps in the Export Process:

- **Selection of Target Data:** Choose the GIS layer(s) or features you want to export.
- **Selection of Output Format:** Choose the desired target format (e.g., Shapefile, KML, GeoPackage, or GeoTIFF).
- **Define Output CRS:** Specify the **Coordinate Reference System** the output file should use. This is often a critical requirement for the recipient of the data.
- **Execution of Export:** The software performs the conversion of the internal data structure and attributes into the selected external format and CRS.

Common Export Scenarios:

- **Sharing Data:** Exporting a layer as a **Shapefile** or **GeoJSON** is the most common way to share vector data.
- **Creating Imagery:** Exporting a raster layer (like a land cover map) as a **GeoTIFF** to be used in image processing software or another GIS.
- **Web Mapping:** Exporting data as **KML/KMZ** for display in Google Earth or as **GeoJSON** for use in web map applications (like Leaflet or OpenLayers).

Challenges in Data Interoperability

- The need for importing and exporting stems from the challenge of **data interoperability**, which is the ability of different systems and software to seamlessly exchange and use information.
- **Format Diversity:** The sheer number of spatial data formats (Shapefile, GeoPackage, GML, GeoTIFF, etc.) necessitates conversion tools.
- **Projection Mismatches:** If the CRS is not correctly identified and transformed, the data will be located incorrectly in the new system (a phenomenon known as **spatial offset** or **misregistration**).
- **Schema Changes:** Different formats and databases have different rules for attribute table structure (e.g., maximum field name length), which can cause data loss or truncation during export.
- Modern GIS software and libraries (like GDAL/OGR) are designed to handle these conversions automatically, making data exchange much smoother than it was historically.

Visualization and Exploration of Spatial Data

- Visualization and exploration are crucial initial steps in spatial data analysis, allowing users to understand data characteristics, discover patterns, and identify potential errors before formal analysis.

Visualization: Cartographic Principles

- **Visualization** is the process of representing spatial data graphically on a map to convey information effectively. Good visualization adheres to core cartographic principles:
- **Symbology:** Using the correct visual variables (size, color, shape, pattern) to represent features and their attributes. For example, using different **colors** to represent categorical land use types or varying **size** to show population density.
- **Classification:** Grouping continuous attribute values into classes to create clearer thematic maps (e.g., natural breaks, equal interval, or quantiles). This transforms raw data into

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- **Scale and Generalization:** Ensuring that the level of detail is appropriate for the map's **scale**. At small scales (showing a large area), features must be **generalized** (simplified or combined) to avoid clutter.
- **Map Elements:** Including essential components like a **title, legend, scale bar, and north arrow** to ensure the map is fully readable and understandable by the audience.

Common Visualization Techniques:

- **Choropleth Maps:** Areas (polygons) are shaded or patterned in proportion to a statistical variable (e.g., average income per county).
- **Proportional Symbol Maps:** A symbol (usually a circle) is placed at a feature's location, and its size is scaled to represent a specific attribute value (e.g., city population size).
- **Dot Density Maps:** Using dots to represent a quantity within an area, where the spatial distribution of the dots conveys the density of the phenomenon (e.g., number of livestock).
- **Heat Maps (Kernel Density Estimation):** Used primarily with point data, these show the intensity or concentration of events by creating a continuous surface, useful for crime or accident analysis.

Exploration: Interactive Data Analysis

- **Exploration** is the interactive process of examining data through visual and statistical methods to gain insights and check data quality. Modern GIS tools facilitate this through:
- **Pan and Zoom:** Interacting with the map by changing the extent (panning) and viewing detail levels (zooming).
- **Attribute Querying:** Clicking on a feature on the map to instantly access its full **attribute data** (tabular information) and verifying its values.
- **Spatial Selection:** Selecting features based on their location, such as finding all features within a specific distance of another feature, or identifying points within a polygon.
- **Statistical Summaries:** Generating basic statistics (mean, median, standard deviation, histograms) for attributes to check the data distribution, identify outliers, and detect errors. For example, a histogram can reveal if an attribute has a normal distribution or if there are unexpected extreme values.
- **Data Linking (Brushing):** The ability to link the map view with an attribute table or chart, so that selecting a feature in one automatically highlights it in the others. This is a powerful technique for identifying the spatial location of statistical outliers.
- Together, effective visualization and exploration are essential for validating data quality and forming the hypotheses that will drive formal spatial analysis.

Chapter 2

Spatial Data Analysis

2.1. Spatial data manipulation and preprocessing

- This chapter covers the core techniques for analyzing spatial data, moving beyond basic visualization to methods that reveal patterns and relationships.

Spatial data (also called geospatial data) includes information about:

- **Location** (coordinates like latitude & longitude)
 - **Shape** (points, lines, polygons)
 - **Relationships in space** (distance, overlap, containment)
- **Data preparation**, particularly the **manipulation and preprocessing of spatial data** is a crucial foundational step before any analysis can commence.
- This multifaceted process involves a range of techniques to **clean** the raw data, ensuring it is **accurate**, **complete**, and **transformed** into a usable format suitable for effective spatial analysis and modeling.

2.1.1 Spatial Data Manipulation

- Spatial data manipulation involves transforming, combining, or modifying geographic features (points, lines, polygons) and their associated attributes to derive new information or structure the data for a specific analysis.

Key Techniques:

1. Buffering: Creates a polygon area around a geographic feature or creates a zone at a specified distance around features (point, line, or polygon) at a specified distance. It's used to identify areas of influence or proximity.

Example: Finding all properties within 100 meters of a river, 500m buffer around a river to identify flood-risk areas

2. Overlay Analysis: Combines two or more spatial datasets to create a new output feature class that contains the geometric and attribute data from the inputs. Common types include:

Common overlay operations:

- **Union:** Preserves all features from the input layers.
- **Intersect:** Outputs only the features (and their attributes) that overlap in all input layers.
- **Erase/Difference:** Removes the area of one input feature from another.
- **Dissolving:** Aggregates features (usually polygons) that have the same value for a specified attribute, merging their boundaries. It simplifies data by reducing the number of features.
- **Example:**
Overlay land-use map with flood zones to find vulnerable residential areas, merging individual county polygons into a single state polygon based on the 'State Name' attribute.

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3. Clipping (Cookie-cutting): Extracts features from an input layer that fall within the boundaries of a specific "clip" feature (usually a polygon). The output keeps the attributes of the input layer.

Example: Isolating streets for a specific neighborhood boundary, clip national road data to show only roads within a specific city.

4. Spatial Join

- Joins attributes from one dataset to another based on spatial relationships or link data by location.

Example: Assign population data (polygon) to schools (points) based on location.

2.1.2 Spatial Data Preprocessing

- **Spatial data preprocessing** focuses on cleaning, correcting, and standardizing raw spatial data to ensure its quality, consistency, and suitability for analysis.
- It is the **first and most important step** in any GIS or spatial analysis project.

Major Steps in Spatial Data Preprocessing

1. Data Cleaning and Validation

- **Handling Missing Data:** Filling gaps in attribute tables (e.g., using mean, median, or spatial interpolation) or dealing with features that lack complete geometry.
- **Error Correction:** Identifying and fixing geometric errors (**topology errors**), such as:

Sliver Polygons: Tiny, unwanted polygons often created during overlay.

Dangles: Lines that don't quite connect where they should (undershoots/overshoots).

Gaps/Overlaps: Polygons that should be adjacent but have small spaces or areas of overlap.

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- **Generalization (Simplification):** Reducing the complexity or detail of features (especially lines and polygons) to improve rendering speed, reduce file size, or make maps more suitable for smaller scales.

Example: Simplifying the highly detailed coastline for a world map.

2. Coordinate System and Projection Management

- **Defining/Assigning Coordinate System:** Ensuring the data has a properly identified Coordinate Reference System (CRS) (e.g., WGS 84, UTM Zone 10N). This is crucial for accurate location and measurement.
- **Reprojection (Transformation):** Converting data from one CRS to another. This is often necessary when combining datasets from different sources.
- **Concept:** Different projections cause different types of distortion (area, shape, distance, or direction). Reprojection standardizes the distortion to a desired measure.

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3. Data Integration and Transformation

- **Geocoding:** Converting non-spatial location descriptions (like street addresses or city names) into spatial coordinates (latitude and longitude).
- **Vector-Raster Conversion:**
- **Rasterization:** Converting vector data (points, lines, polygons) into a raster format (grid cells).
- **Vectorization:** Converting raster data (like a scanned map or satellite image classification) into vector features.
- **Attribute Join/Relate:** Linking spatial features to non-spatial data tables based on a common key field to enrich the geographic data.

2.2. Point Pattern Analysis

- **Point Pattern Analysis (PPA)** is a branch of **spatial statistics** that focuses on studying the spatial arrangement of discrete point events within a defined geographic region and used to study the **distribution of point features** (events or objects) in a geographic space.

These points could represent phenomena such as:

- Locations of trees in a forest
 - Crime incidents in a city
 - Disease cases in a region
 - Retail store locations
- The core goal is to determine if the pattern of points is random, clustered, or dispersed, and to infer the underlying **spatial process** that generated that pattern.

Core Concepts

- **Point Pattern:** A dataset of locations (coordinates) of "events" that could occur at many places but only happen in a few, such as disease cases, crime incidents, tree locations, or retail store locations.
- **Clustering** refers to a spatial pattern in which **points or events tend to occur close together** rather than being evenly or randomly spaced. It indicates that the presence of one event **increases the likelihood** of another occurring nearby.
- **Study Region (Window, D):** The defined boundary within which the point pattern is observed and analyzed. The boundary's definition is crucial as it affects all statistical calculations.
- **Intensity (λ):** The expected number of points per unit area.
$$\lambda = \frac{n}{A}$$
- where n = number of points, A = area.

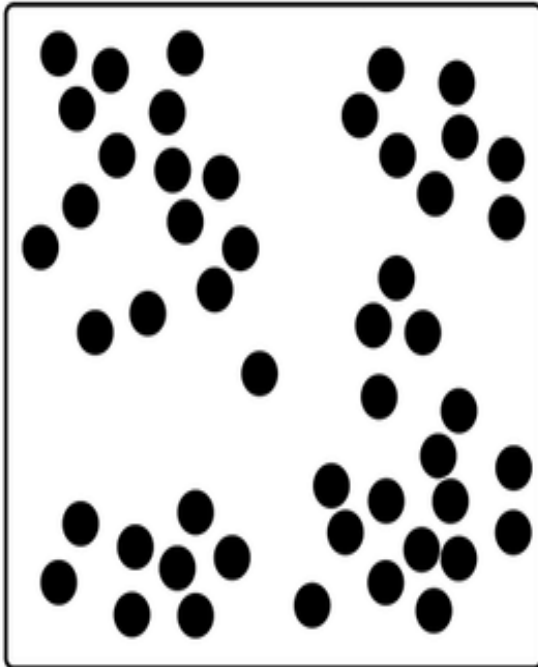
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- **Complete Spatial Randomness (CSR):** The **null hypothesis** in PPA. It represents a theoretical process where:
 - Any point is equally likely to occur at any location within the study region.
 - The location of one point does not influence the location of any other point (spatial independence).

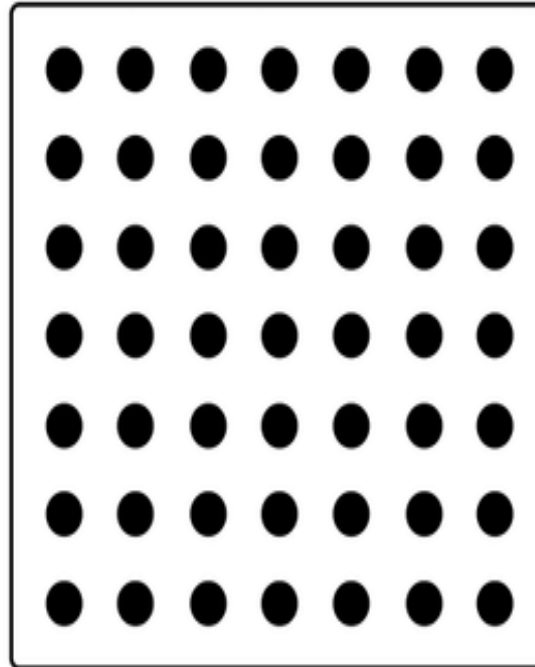
Types of Patterns

1. PPA seeks to distinguish the observed pattern from CSR:
 - **Clustered:** Points appear grouped together, indicating an **attractive** process (e.g., disease spreading via contagion, stores locating near a central market, crime hotspots).
 - **Dispersed (Regular/Uniform):** Points are more regularly spaced than random, indicating a **repulsive** or competitive process (e.g., individual tree territories, evenly distributed service facilities).
 - **Random:** The pattern is consistent with CSR. Points occur with no predictable pattern. Each location has an equal probability of containing a point.

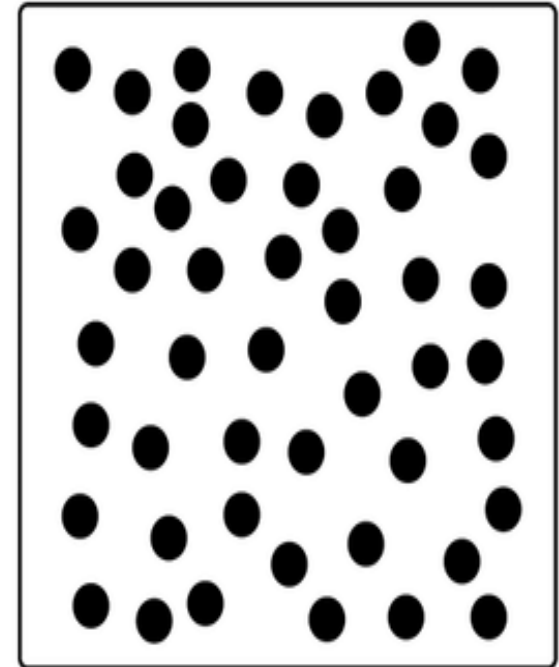
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Clustered



Regularly dispersed



Random

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Orders of Point Properties

- The **orders of point properties**, which describe different levels of **spatial dependence or structure** in a point process. Point patterns are analyzed based on

Property Order	Focus	Measure	Description
First-Order	Intensity/Density	Mean Center, Kernel Density	Focuses on the variation in the average number of points per unit area across the study region (inhomogeneity). It is determined by external factors (covariates) like elevation or population density.
Second-Order	Interaction	Nearest Neighbor, Ripley's function K	Focuses on the relationship between the points themselves (attraction or repulsion), independent of external factors. It measures clustering or dispersion.

2.2.1 Density-Based Methods (First-Order)

- These methods measure the variation in the **intensity** (λ) of the point process across the study area.

2.2.1.1. Quadrat Analysis

- **Quadrat Analysis:** Divides the study region into a regular grid (quadrats) and counts the number of points in each cell. The variance-to-mean ratio (or Index of Dispersion) of the counts is used to test for CSR.

- Variance-to-mean ratio (VMR) is computed:

$$VMR = \frac{Variance}{Mean}$$

- **Interpretation:**
- $VMR=1 \rightarrow$ random
- $VMR>1 \rightarrow$ clustered
- $VMR<1 \rightarrow$ regular

Limitation: Highly sensitive to the size and placement of the grid.

2.2.2 Distance-Based Methods (Second-Order)

- **Nearest Neighbor Analysis (NNA):** Compares the **average distance** from each point to its nearest neighbor (observed) against the average distance expected under CSR. The **Nearest Neighbor Ratio or Index of Cluster/Dispersion(R)** is calculated as :

$$R = \frac{r_o}{r_e}$$

Where r_o = observed mean distance

$$r_e = \text{expected mean distance}; r_e = \frac{1}{2\sqrt{(n/A)}}$$

- n = number of points
- A = area of the region (same units as distances)
- **Interpretation:**
- $R \approx 1$: Random
- $R < 1$: Clustered
- $R > 1$: Dispersed
- **Limitation:** Only uses the single closest distance, ignoring interactions at larger scales.

2.2.2.2. Ripley's K Function

- Ripley's K Function is a **second-order spatial statistic** used to analyze the **spatial distribution of point patterns** over a range of distances.
- It measures the expected number of neighbors within a given distance **d** of any point, and compares it to the expected count under CSR.
- Unlike the Nearest Neighbor or Quadrat methods (which use a single scale), Ripley's K examines clustering or dispersion **at multiple spatial scales** and Provides more detailed insight than NNA or Quadrat methods.
- It plots $K(d)$ against distance d , allowing for analysis of clustering or dispersion at **multiple scales**
- It helps to answer: *“At what distances do points show clustering or regularity?”*

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- For a study region **A** with **n points**, Ripley's K function is defined as:

$$K(d) = \frac{A}{n^2} \sum_i \sum_{j \neq i} I(d_{ij} \leq d)$$

Where: A = area of the study region

n = total number of points

d_{ij} = distance between point *i* and *j*

I(d_{ij} ≤ d) = indicator function (1 if distance ≤ *d*, otherwise 0)

- Expected K for Complete Spatial Randomness (CSR)
- If the pattern is **completely random**, the expected value of K(d) is: $K_{CSR}(d) = \pi d^2$

Interpretation:

- If $K(d) > K_{Poisson}(d)$: clustering at distance *d*.
- If $K(d) < K_{Poisson}(d)$: dispersion at distance *d*.
- If $K(d) \approx K_{Poisson}(d)$: Random pattern at distance *d*.

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L(d) Transformation (Linearized Form)

- Often the **L(d) function** (a variance-stabilized transformation of K(d)) is plotted to simplify interpretation against a straight line.
- Plotted as **L(d)** for interpretability:

$$L(d) = \sqrt{\frac{K(d)}{\pi}} - d$$

- **Interpretation:**
- $L(d)=0 \rightarrow$ Random
- $L(d)>0 \rightarrow$ Clustering
- $L(d)<0 \rightarrow$ Regularity
- Plotting **L(d) vs. d** helps visualize how the pattern changes across scales.

Example

- **Bike Theft Locations in a City Park**
- Imagine a city park area of $1 \text{ km} \times 1 \text{ km}$ (so $A=1,000 \text{ m} \times 1,000 \text{ m}=1,000,000 \text{ m}^2$). The record a locations (in meters) of **5 bike thefts**:

Theft	X	Y
1	100	200
2	120	220
3	130	210
4	800	900
5	810	910

Soln.

Step1: Compute Distances

- Theft 1, 2, 3 are **close together**, while 4 and 5 are **far away** (but close to each other).
- Example distances (in meters): Distances between theft locations are calculated using the **Euclidean distance formula** (straight-line distance). For two points: (x_1, y_1) and (x_2, y_2) the distance is:

$$d = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$$

- Between (1,2): 28.3
- Between (1,3): 31.6
- Between (4,5): 14.1
- Between (1,4): 989.9 → far apart

Step2: Choose a distance scale d

- Let's test for $d=50$ m, count all pairs that are **within 50 meters**.
- Pairs within 50 m: (1,2), (1,3), (2,3), (4,5) → **4 pairs**
- Total possible pairs = $n(n-1)=5 \times 4=20$

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Step3: Compute $K(50)$

$$K(50) = \frac{A}{n^2} \sum I(d_{ij} \leq 50)$$

$$K(50) = \frac{1,000,000}{25} \times 4 = 160,000$$

Step4: Compare to CSR (Complete Spatial Randomness)

$$K(d) = \pi d^2$$

$$K_{CSR}(50) = \pi(50)^2 = 7,854$$

- Under CSR (random pattern), we expect:
- Since $K_{obs}(50) = 160,000 > 7,854 \rightarrow$ **strong clustering**

Distance (m)	K(d) (observed)	K(d) under CSR	Pattern
50	160,000	7,854	Clustered
100	165,000	31,416	Clustered
500	800,000	785,398	~Random

2.2.3. Applications

PPA is widely used in numerous fields:

- **Epidemiology:** Identifying **disease clusters** (e.g., cancer, cholera, as in John Snow's famous 1854 work) to pinpoint sources of infection or environmental risk factors.
- **Ecology:** Analyzing the distribution of **plant or animal species** to infer ecological processes like competition (leading to dispersion) or resource attraction (leading to clustering).
- **Criminology:** Mapping **crime hotspots** to allocate police resources effectively and understand the spatial behavior of criminal activity.
- **Urban Planning/Retail:** Studying the distribution of **businesses or services** to determine optimal facility locations or assess market competition.

2.2.4 Bivariate and Space-Time Point Patterns

- **Bivariate and Space-Time Point Patterns** are extensions of **Point Pattern Analysis (PPA)**, a set of statistical techniques used to analyze the locations of events or objects (points) occurring within a bounded region, aiming to characterize the underlying spatial process.

2.2.4.1 Bivariate Point Patterns

- A **Bivariate Point Pattern** involves **two different types of events or point sets** occurring within the same spatial region. This is a type of **multivariate** or **marked** point pattern where the mark is a binary or categorical variable differentiating the two types.
- **Bivariate analysis** examines the relationship between two different point patterns (e.g., crime locations and streetlights) to see if one influences the other's distribution.
- **Focus:** The primary goal is to analyze the **spatial relationship** between the two types of points.

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Key Questions: Are the two types of points:

- **Clustered** together (attraction)? For example, do cases of a certain disease tend to occur near a specific environmental factor location?
- **Dispersed** relative to each other (repulsion/competition)? For example, do different tree species repel each other?
- **Independent** of each other?
- **Analysis:** Methods often involve extending univariate PPA techniques, such as distance-based functions (e.g., the **cross K-function** or **cross pair correlation function**), to measure the interaction between the two sets of points across different spatial scales.

Difference from Univariate PPA

Univariate

Measures clustering within same
type of event

Uses $K(r)$

Why It's Important

Bivariate PPA helps answer **interaction questions**, such as:

- Do facilities reduce crime nearby?
- Do certain species compete?
- Are pollution sources influencing disease clusters?

Bivariate

Measures interaction between
two types

Uses $KAB(r)$

Methods of Analysis

a. Cross K Function (Ripley's $K_{12}(d)$)

- The **Cross K Function**, denoted as $K_{12}(d)$, is an extension of **Ripley's K Function** for **bivariate point patterns**; that is, when you have **two different types of points** (e.g., species A and species B, or schools and parks).
- It measures the **spatial relationship between two point types over different distance scales (d)** to determine whether the two types are:
- **Attracted (clustered together)**
- **Independent (randomly located)**
- **Repelled (segregated)**

Purpose

- To answer questions like:
- Do **species A** trees tend to occur near **species B** trees?
- Are **crimes of type 1** located close to **crimes of type 2**?
- Do **restaurants** and **bars** form mixed clusters in the city?

A generalization of Ripley's K-function for two point types (Type 1 and Type 2).

$$K_{12}(d) = \frac{A}{n_1 n_2} \sum_i \sum_j I(d_{ij} \leq d) w_{ij}^{-1}$$

Where:

- A: area of the study region
- n_1, n_2 : number of points of type 1 and type 2
- $I(d_{ij} \leq d)$: indicator function (1 if distance $\leq d$, else 0)
- W_{ij} Edge correction weight (to adjust for boundary effects)

Expected Value Under Independence

- If the two point types are **spatially independent**, the expected value is: $K_{CSR}(d) = \pi d^2$
- This is the same as for complete spatial randomness (CSR).

Cont...

Interpretation:

- $K_{12}(d) > K_{CSR}(d)$: attraction between types.
- $K_{12}(d) < K_{CSR}(d)$: repulsion.
- $K_{12}(d) = K_{CSR}(d)$: spatial independence.

The $L_{12}(d)$ form is often used for easier interpretation:

$$L_{12}(d) = \sqrt{\frac{K_{12}(d)}{\pi}} - d$$

- **Interpretation:**
- $L_{12}(d) > 0$: Attraction
- $L_{12}(d) = 0$: Independence
- $L_{12}(d) < 0$: Repulsion

Example

Case: You study two species of trees:

- Type 1 = Acacia
- Type 2 = Eucalyptus
- After computing $K_{12}(d)$:

Distance (m)	$K_{12}(d)$	πd^2	$L_{12}(d)$	Interpretation
5	90	78.5	+1.5	Attraction
10	320	314	+0.4	Independence
20	1100	1256	-1.3	Repulsion

Interpretation:

- At short distances, **A and B are clustered** together.
- At larger distances, they become **segregated**.

b. Nearest Neighbor Cross-Type Analysis

- Nearest Neighbor Cross-Type Analysis (NNCT Analysis) is a method used in bivariate point pattern analysis to study spatial interaction between two types of points.
- It examines whether points of one type tend to occur near points of another type, helping to detect attraction (association) or repulsion (segregation) between them.

Purpose: To answer questions such as:

- Do Type A points (e.g., schools) tend to locate near Type B points (e.g., parks)?
- Are Species A trees more likely to be found near Species B trees?
- Do violent crimes occur close to property crimes?

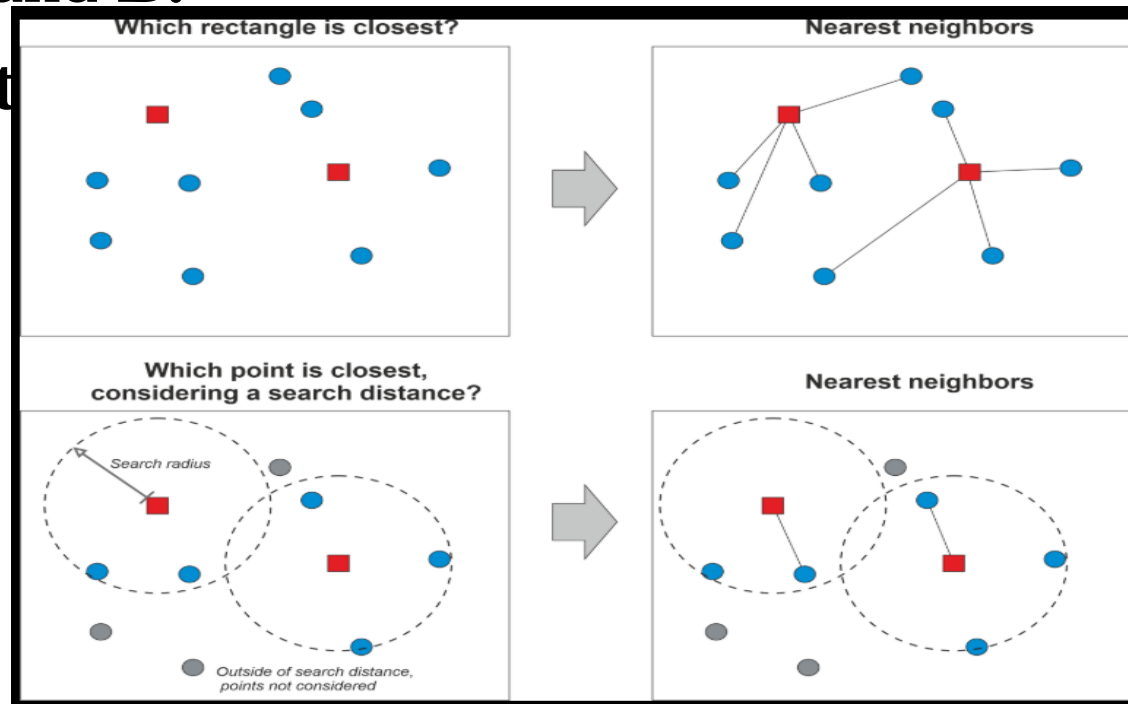
Basic Concept: For each point of one type (say, A), we find the nearest neighbor of another type (say, B) and measure the distance between

them.

Applications

- **Ecology:** Interaction between species (competition or symbiosis).
- **Urban Geography:** Relationship between facilities (e.g., schools and **B.**

Visualizat



Cont...

Steps in Nearest Neighbor Cross-Type Analysis

Step 1: Identify Point Types

- Label your dataset with two distinct categories:
- Type **A** (e.g., Schools)
- Type **B** (e.g., Hospitals)

Step 2: Compute Nearest Neighbor Distances

- For each point of type A, find the **nearest point of type B**.
- Measure the **Euclidean distance** ($d_{a\beta}$) between them.

Cont...

Step 3: Calculate Mean Nearest Neighbor Distance

$$r_{AB} = \frac{1}{n_A} \sum_{i=1}^{n_A} d_{i(AB)}$$

Where:

- r_{AB} : observed mean distance from A to nearest B
- n_A : number of A-type points
- $d_{i(AB)}$: distance from A-point i to its nearest B-point

Step 4: Compare with Expected Distance Under Randomness

- Expected mean distance if A and B are spatially independent:

$$r_{AB}^{(exp)} = \frac{1}{2\sqrt{\lambda_B}}$$

- Where $\lambda_B = \frac{n_B}{A}$ is the density of type B points per unit area.

Cont...

Step 5: Compute Cross-Type Index (R_{AB})

$$R_{AB} = \frac{r_{AB}}{r_{AB}^{(exp)}}$$

- Interpretation

Value of R_{AB}	Interpretation	Spatial Relationship
$R_{AB}=1$	Random distribution	Independence
$R_{AB}<1$	A is closer to B than random	Attraction / Association
$R_{AB}>1$	A is farther from B than random	Repulsion / Segregation

- You can also perform the reverse (B-to-A) to check **directional association** (asymmetry).

Example

- **Study:** Locations of coffee shops (A) and bookstores (B) in a city. 20 Coffee shops (Type A), 15 Bookstores (Type B) and Study area = 4 km² downtown district

For each coffee shop, measure the distance to the **nearest bookstore** the distances (meters): 120, 85, 140, 95, 110, 75, 130, 100, 90, 105, 80, 115, 125, 70, 135, 95, 100, 85, 110, 90.

Soln:

- Calculate Mean Nearest Neighbor Distance is approximately: $r_{AB} = \frac{1}{n_A} \sum_{i=1}^{n_A} d_{i(AB)} = 103$
- Expected mean distance: $r_{AB}^{(exp)} = \frac{1}{2\sqrt{\lambda_B}} = 1/(2\sqrt{15/4}) = 0.258 \text{ km} = 258 \text{ meters}$

Compute Cross-Type Index (R_{AB})

$$R_{AB} = \frac{r_{AB}}{r_{AB}^{(exp)}} = 103/258 \approx 0.39$$

- □ Since $R_{AB} < 1$, **coffee shops tend to be near bookstores** → spatial attraction.

2.2.4.2 Space-Time Point Patterns

- A **space-time (spatio-temporal) point pattern** is a collection of events where each event is recorded not only by its **spatial location** (s) but also by its **time of occurrence** (t). The data forms a set of points $\{(s_i, t_i)\}$ within a spatio-temporal domain.
- **Space-time analysis** incorporates a temporal dimension, looking for patterns in how point locations change over time (e.g., how the locations of a disease outbreak spread over several weeks).
- **Focus:** The analysis characterizes how the pattern unfolds over both **space** and **time**, accounting for potential **spatial**, **temporal**, and **spatio-temporal interactions**.

Cont...

Key Questions:

- Is there **temporal clustering** (e.g., bursts of events) that occurs across the study area?
- Is there **spatial clustering** that persists or changes over time?
- Is there **spatio-temporal clustering**, indicating that an event at a certain location and time increases the likelihood of another event occurring nearby in both space and time (a phenomenon known as **self-excitement** or **contagion**)? For example, crime incidents or disease outbreaks.

Example:

- Daily crime events in a city.
- Spread of a disease outbreak.
- Earthquake occurrences over months or years.

Objectives

- To detect **spatial clustering** over time.
- To identify **spatio-temporal hotspots** (areas active during specific times).
- To understand **diffusion processes** (how events spread through space and time).

Cont...

- **Analysis:** This typically involves analyzing the **intensity function** $\lambda(s,t)$ to look for trends and using spatio-temporal extensions of second-order statistics, such as the **space-time K-function** or **pair correlation function**, which measure how the expected number of event pairs changes with both spatial distance and time difference.
- Space-Time K Function ($K(s, t)$)
- The **Space-Time K Function**, often denoted as $K(s,t)$, is a statistical tool used in **spatial-temporal point pattern analysis**. It extends Ripley's K-function (which is purely spatial) by incorporating the **temporal dimension**, allowing us to study **clustering or dispersion of events** over both space and time.

$$K(s,t) = \frac{E(\text{number of other events within distance } s \text{ and time } t)}{\lambda}$$

where:

- λ = overall space-time intensity (average number of events per unit area per unit time),
- s = spatial distance,
- t = temporal distance.

Cont...

Interpretation:

- If events occur **independently** in space and time (a *spatio-temporal Poisson process*):
- $K_{CSR}(s,t) = \pi s^2 t$
- $K(s,t) > K_{CSR}(s,t)$: space-time clustering.
- $K(s,t) < K_{CSR}(s,t)$: space-time regularity.

Practical Example: Bike Theft Data in a City

- Let's imagine you're a data analyst working for the police in **Cityville**.
- We have data on **bike thefts** with:
- **Location (x, y)** — the coordinates of where the theft happened
- **Time (t)** — when it happened (e.g. in days since the start of the year)

Cont...

Theft ID	X (m)	Y (m)	Time (days)
1	100	200	10
2	110	210	11
3	300	800	12
4	115	220	13
5	310	820	40

Step 1: Visualize

- Plot the thefts on a city map and time line.
You'll see that thefts 1, 2, and 4 are close **in both space and time**, while 3 and 5 are far apart.

Step 2: Compute $K(s,t)$

- we define: $s=50$ meters (spatial neighborhood), $t=5$ days (temporal neighborhood)
- Now we need: On average, how many other thefts occurred within **50 m** and **5 days** of a given theft? And the area is $210 \times 620 = 130,200 \text{ m}^2$ and time window is 30 days.
- We need count for each theft how many others fall within that space–time window, average over all thefts, and adjust for overall density.

Cont..

- There are 10 possible pairs ($n(n-1)/2 = 5 \times 4/2 = 10$).
- Euclidean spatial distance **Pair (1,2)**
- Spatial distance: $\sqrt{(110 - 100)^2 + (210 - 200)^2} \approx 14.14$
- Absolute time difference: $|11 - 10| = 1$ day
- Close in space ($\leq 50\text{m}$): (1,2), (1,4), (2,4) and (3,5)
- Now check time (≤ 5 days): (1,2), (1,4) and (2,4)
- $\lambda = n/(A * t) = 5/(130,200 * 30) = 1.28 \times 10^{-6}$
- Expected = $6/5 = 1.2$
- $K(50,5) = 1.28 \times 10^{-6} / 1.2 = 937,500$

Cont...

Step 3: Interpret

- You compare your empirical $K(s,t)$ to what you'd expect under **complete spatial–temporal randomness (CSTR)**.
- Expected under space–time independence:
- $KCSR(s,t) = \pi s^2 t = \pi(50^2)(5) = 39,270$ Since
- $937,500 > 39,270$ there is **strong space–time clustering**, driven by thefts 1–2–4 occurring close together in both space and time.

s (m)	t (days)	Observed $K(s,t)$	Expected $K(s,t)$	Interpretation
50	5	Higher	Lower	Space–time clustering — thefts happen near each other in both space and time
50	20	Similar	Similar	Random in time when the window is larger
500	5	Similar	Similar	Not much large-scale spatial clustering

- So, we conclude: “Bike thefts tend to cluster locally in space and time ; possibly due to repeat offenders targeting the same area over a few days.”

2.3. Lattice (Grid) and Image (Raster) Data Analysis

- Lattice or raster data represent **spatially continuous phenomena**, such as temperature, elevation, population density, or land cover.
- Both **lattice (grid)** and **image (raster)** data represent spatial information using a **regular grid** of cells (pixels). Each cell has a specific **spatial location** (x, y) and an associated **attribute value**.
“lattice,” “grid,” and “raster” are conceptually similar, differing mainly by **discipline and context**.

2.3.1 Lattice (Grid) Data

- **Lattice (or Grid) data** are spatial data collected at **regularly spaced locations** arranged in a grid pattern. Each grid cell (square or rectangle) has a **single measured or estimated value** representing the attribute for that area.

Examples:

- Population density per administrative unit.
- Average temperature in each 10×10 km grid cell.
- Air pollution index at monitoring grids.

Structure

- Lattice/Grid Example

```
-----  
|12|13|15|14|16|  
|11|12|14|13|15|  
|10|11|13|12|14|  
| 9|10|12|11|13|  
-----
```

- Each cell = data value
- The grid cells form a **regular lattice** (like a chessboard).
- Each cell has a **coordinate (row, column)** and an associated **attribute value**.

Types of Lattice Data

a. Regular Lattice: The spatial units are uniform shapes and sizes, typically **square** (forming a grid or **raster**), or sometimes **hexagonal** or **triangular**. This is common in remote sensing imagery (pixels) or climate model outputs.

Cont...

b. Irregular Lattice (Areal Data): The spatial units are heterogeneous in size and shape, often defined by administrative, political, or census boundaries, such as counties, tracts, or zip codes. Data like cancer rates by county is a classic example.

Analytical Methods

- The main goal of lattice data analysis is to account for and model **spatial autocorrelation**—the tendency for observations closer in space to be more alike than those farther apart.

Visualization

- **Choropleth maps** for areal data.
- **Contour maps** for continuous fields.
- **Heatmaps** for grid intensity patterns.

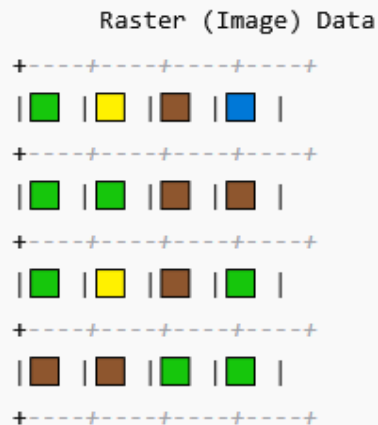
Cont...

Method	Purpose	Example Application
Spatial Autocorrelation (Moran's I, Geary's C)	Measures similarity of neighboring cells	Detecting clustering in temperature
Local Indicators of Spatial Association (LISA)	Identifies local hotspots and cold spots	Disease risk mapping
Spatial Regression	Models relationships with spatial dependence	Linking air quality to traffic density
Interpolation (Kriging, IDW)	Predicts values at unsampled locations	Estimating rainfall surface
Trend Surface Analysis	Fits mathematical surfaces to data	Modeling topographic variation

2.3.2 Image (Raster) Data

- **Raster (image) data** represent spatial phenomena as a grid of **pixels (cells)**, where each pixel stores a **numeric value** (e.g., brightness, temperature, elevation).
 - A raster dataset is essentially a matrix (or array) of equally-sized square grid cells, or **pixels**, that cover a given area.
- remote sensing and GIS.**

Structure



Cont...

Each cell = pixel (reflectance/intensity)

- The raster grid is composed of **rows and columns of pixels**.
- Each pixel corresponds to a specific geographic area.
- Pixel value = measurement of some property (e.g., reflectance, NDVI, elevation).
- ☐ **Tip:** Use color pencils, markers, or fill tool to color each pixel differently.
 -  = Water
 -  = Vegetation
 -  = Urban/Bare soil
 -  = Farmland

Types of Raster Data

Type	Description	Example
Continuous Raster	Values change smoothly across space	Elevation, temperature
Categorical Raster	Represents discrete classes	Land cover (forest, water, urban)
Multiband Raster	Several layers (bands) for the same area	Satellite imagery (RGB, infrared)

Analytical Methods

Method	Description	Application
Raster Algebra (Map Algebra)	Combine layers using arithmetic or logical operations	Suitability analysis (e.g., slope + soil + land cover)
Filtering (Convolution)	Smooth or enhance features using moving window	Edge detection in images
Classification	Group pixels into categories	Land cover mapping
Change Detection	Compare rasters from different times	Deforestation monitoring
Texture Analysis	Quantifies spatial arrangement of pixels	Urban vs. rural land analysis
NDVI (Normalized Difference Vegetation Index)	$(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$	Vegetation health assessment

Cont...

Visualization

- Display as **grayscale** (one band) or **color composite** (multiple bands).
- Create **heatmaps** or **3D surface models**.
- Overlay with vector data (roads, boundaries).

Chapter 3

Geostatistical Analysis of Spatial Data

3.1. Introduction to geostatistics

- The term “geostatistics” was coined by Matheron (1962, 1963) to describe the statistical methodology for examining ore reserves from spatially distributed data in an ore body.
- Geostatistics is a branch of spatial statistics that focuses on analyzing and modeling spatially continuous data.
- It is used to describe, predict, and simulate values of a variable distributed across geographic space (e.g., soil moisture, pollution concentration, mineral content).
- It is widely used when measurements are taken at specific locations, but we want to estimate values in places where no measurements exist.

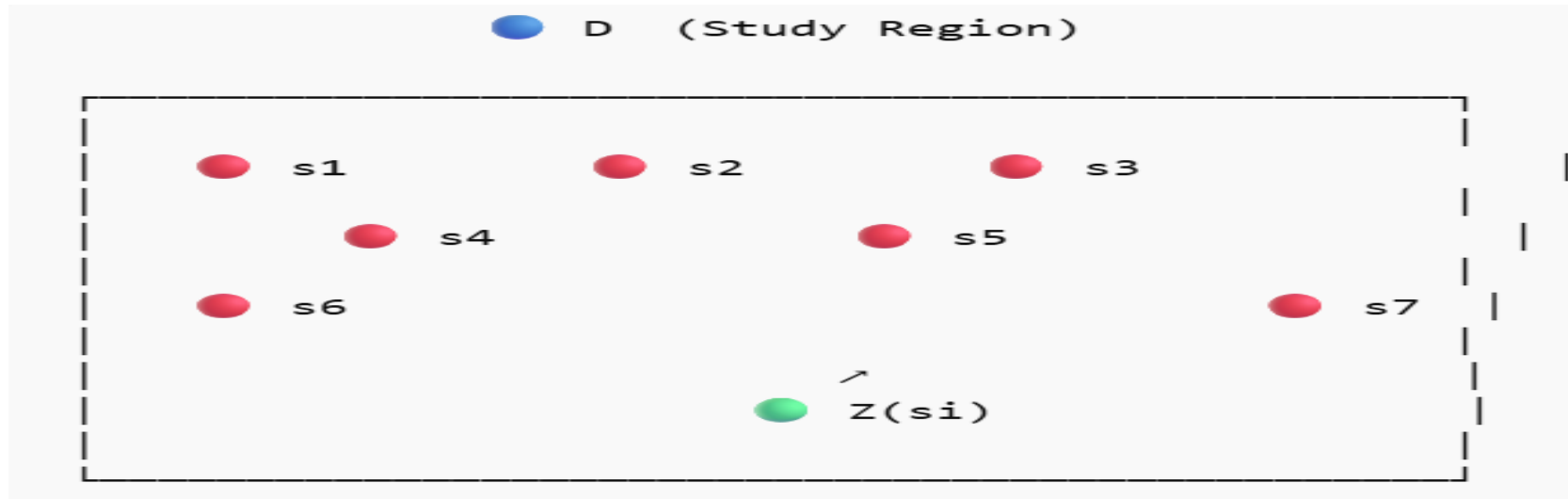
Cont...

- Let $Z(s_1), \dots, Z(s_n)$ be observations of a spatial variable Z at locations $s_1, \dots, s_n$. In many situations, geostatistical data may be assumed to be a partial realization of a random process.

$$\{Z(s): s \in D \subset \mathbb{R}^2\},$$

- Where $Z(s)$ = a random variable at location s , D = the spatial region being studied, $\mathbb{R}^2$ = 2-dimensional space (x and y coordinates)
- Based upon this partial realization, we may seek to infer the characteristics of the spatial process that gives rise to the data observed such as the mean and variability of the process.
- Then, we can use this information to predict the process at unsampled locations and construct a spatially continuous surface of the variable of study.

Cont..



Color Meaning

-  $D \rightarrow$ Study region (spatial domain)
-  Points ($s_1, s_2, \dots, s_n$) $\rightarrow$ observation locations
-  $Z(s_i) \rightarrow$ value of the spatial variable measured at location s_i

Key assumptions:

- Spatial data exhibit spatial dependence nearby locations tend to have similar values.
- This spatial dependence can be quantified and modeled for interpolation or prediction (e.g., kriging).

Purpose and Scope

- The main goal of geostatistics is to quantify spatial patterns and predict spatial distributions of variables. It helps answer questions like:
 - How similar are values at nearby locations?
 - How far does spatial correlation extend?
 - What is the best estimate of a variable at an unmeasured location?

3. 2. Spatial Autocorrelation

- Spatial autocorrelation refers to the degree to which the value of a variable at one location is similar (or dissimilar) to values at nearby locations. It measures the spatial dependence among observations distributed across a geographic space.

Type of Spatial Autocorrelation

- Positive autocorrelation: similar values cluster together (e.g., high rainfall areas near other high rainfall areas).(e.g., high rainfall areas surrounded by other high rainfall areas).
- Negative autocorrelation: dissimilar values are neighbors (e.g., high values next to low ones).(e.g., a high population density area surrounded by low-density regions).
- No autocorrelation: random spatial distribution.

Importance of Spatial Autocorrelation

Spatial autocorrelation helps us to understand spatial patterns and processes. It is essential for:

- Detecting **clustering or dispersion** of geographic features.
- Assessing the **spatial structure** of data before modeling (e.g., kriging).
- Ensuring valid statistical inference (traditional statistics assume independence, which spatial data often violate).

Measure of Spatial Autocorrelation

- Spatial autocorrelation can be assessed using indices that summarize the degree to which similar observations tend to occur near each other over the study area.
- Two common indices that are used to assess spatial autocorrelation in areal data are Moran's I and Geary's C.

Moran's I

- Moran's I is a statistic used to measure spatial autocorrelation that is, how similar or dissimilar values are across space. It tells us whether nearby locations tend to have similar values (clustering), dissimilar values (dispersion), or no spatial pattern (randomness).

What Moran's I Measures

- **Positive Moran's I** ($I > 0$) → nearby locations have **similar** values (spatial clustering).
- **Negative Moran's I** ($I < 0$) → nearby locations have **dissimilar** values (spatial dispersion/alternation).
- **Near zero** ($I = 0$) → no spatial autocorrelation (random spatial pattern).

Key Components

To calculate Moran's I, three things are needed:

- **The Variable of Interest (Z_i):** The attribute value for each location i .
- **Deviations from the Mean ($D_i = Z_i - \bar{Z}$):** The difference between the value at each location and the mean value for all locations.
- **A Spatial Weights Matrix (w_{ij}):** A matrix that defines how spatial units are related or "neighborly." This can be based on distance (e.g., within 10 km), contiguity (e.g., sharing a border), or other criteria.

$$I = \frac{n}{W} * \frac{\sum_i \sum_j (z_i - \bar{z})(z_j - \bar{z})}{\sum_i (z_i - \bar{z})^2}$$

Cont...

Where:

- n : number of spatial units
- w_{ij} : spatial weight between locations i and j
- Z_i, Z_j : observed values at locations i and j
- $\bar{Z}$: mean of the variable
- $W = \sum_i \sum_j w_{ij}$: sum of all weights

Interpretation and Significance

- Moran's I is an inferential statistic, meaning the index value alone is not enough; you must also evaluate its statistical significance using a Z -score and p -value.
- The Null Hypothesis (H_0) for Moran's I is that the attribute being analyzed is randomly distributed among the features in the study area.

Cont...

- If the p-value is small (e.g., <0.05), you reject the null hypothesis, concluding that the pattern is *not* random.
 - ✓ A positive Z-score with a small p-value indicates significant clustering (positive autocorrelation).
 - ✓ A negative Z-score with a small p-value indicates significant dispersion (negative autocorrelation).
- If the p-value is large (e.g., ≥ 0.05), you fail to reject the null hypothesis, concluding that the pattern is statistically random.

Global vs. Local Moran's I

- Global Moran's I: Provides a single, overall measure of spatial autocorrelation for the entire study area. It tells us if the data is clustered/dispersed, but not where the clusters/outliers are located.
- Local Moran's I (LISA - Local Indicators of Spatial Association): Calculates a separate Moran's I value for each individual location. This allows you to identify:

Cont...

- **Hot Spots (High-High):** Areas of high value clustered with other high values.
- **Cold Spots (Low-Low):** Areas of low value clustered with other low values.
- **Spatial Outliers (High-Low or Low-High):** An area with a high value surrounded by low-value neighbors, or vice versa.

Example: Moran's I in Public Health (Disease Mapping). Scenario: Mapping COVID-19 Incidence

- Imagine a public health department wants to know if high rates of COVID-19 incidence are clustered in specific counties of a state, or if they are randomly scattered.
- **Variable of Interest (x_i):** The rate of new COVID-19 cases per 100,000 residents for each county.

Cont...

- **Spatial Units (n):** The individual counties.
- **Spatial Weights (w_{ij}):** Usually based on **contiguity** counties that share a physical border are considered neighbors.

Application of Global Moran's I

- The health department calculates the Global Moran's I for the whole state.

Calculated Value	Conclusion	Interpretation
$I = +0.65$	Significant Positive Autocorrelation (Z-score > 1.96, $p < 0.05$)	The high rates of COVID-19 are not random. Counties with a high incidence rate tend to be immediately next to other counties with high incidence rates (Hot Spots), and low-rate counties tend to be next to low-rate counties (Cold Spots).

Cont...

- Implication for Policy: This finding suggests that spatial processes are at work.
- Factors like population density, access to healthcare, shared economic activity, or common travel patterns are likely causing the disease to spread across neighboring.

Application of Local Moran's I (LISA)

- After finding overall clustering (Global I = +0.65), the department uses **Local Moran's I** to pinpoint the exact locations of concern.

The LISA map identifies four key areas:

1. **High-High Clusters (Hot Spots):** A cluster of inner-city counties showing the highest rates.
 - Action: Prioritize vaccine distribution, mobile testing centers, and enhanced masking mandates here.
2. **Low-Low Clusters (Cold Spots):** A cluster of rural, isolated counties showing the lowest rates.
 - Action: Maintain standard protocols, but monitor for incoming cases from neighboring regions.
3. **High-Low Outliers:** A single county with a very high rate, completely surrounded by counties with low rates.
 - Action: Investigate a local, non-spatial cause, such as a single outbreak event (e.g., a large factory or university campus) or a reporting error.
4. **Low-High Outliers:** A single county with a very low rate, surrounded by counties with high rates.
 - Action: Investigate the low-rate county to see if there are best practices (e.g., high early vaccination rates, a successful localized public service campaign) that could be exported to the neighboring "Hot Spot" counties.

Moran's I in Economics (Housing Prices)

Scenario: Analyzing Housing Price Inflation

- An urban planner wants to know if high housing prices in one neighborhood are influencing (or are being influenced by) the prices in adjacent neighborhoods in a large metropolitan area.
- **Variable of Interest (x_i):** The average sale price of homes in a specific census tract or neighborhood.
- **Spatial Units (n):** The census tracts or neighborhoods.
- **Spatial Weights (w_{ij}):** Could be based on a fixed distance threshold (e.g., all neighborhoods whose centers are within 2 miles of each other are neighbors).

Cont...

- The planner calculates the **Global Moran's I** for the entire city.

Calculated Value	Conclusion	Interpretation
$I = +0.81$	Very Strong Positive Autocorrelation (Z-score significantly high, $p < 0.01$)	This indicates that housing prices are highly clustered. Expensive neighborhoods tend to border other expensive neighborhoods, and cheaper neighborhoods tend to border other cheaper neighborhoods.

- Implication for Policy:** The finding confirms the presence of "Neighborhood Effects." Factors driving up prices (e.g., school quality, access to transit, zoning restrictions) are not isolated; they are spilling over and concentrating wealth or poverty spatially. This suggests city-wide planning and zoning policies have a strong localized impact.

Application of Local Moran's I (LISA)

- Since the Global Moran's I showed strong clustering, the planner maps the **Local Moran's I** results to inform targeted development strategies.
- The LISA map helps pinpoint specific spatial relationships:
 - 1. High-High Clusters (Hot Spots):** The inner core and select affluent suburbs with persistently high housing values.
- **Economic Interpretation:** These are areas of high demand and constrained supply.
- **Action:** The city might explore increasing density (e.g., allowing multi-family housing) or taxing land value more heavily here to capture market growth for public services.

Cont...

2. Low-Low Clusters (Cold Spots): Areas farther from the central business district or along industrial corridors where prices are consistently low.

- **Economic Interpretation:** These neighborhoods face systemic disadvantages.
- **Action:** Target investments here for infrastructure improvements, affordable housing construction, or economic development zones to raise property values and opportunity.

3. High-Low Outliers: An expensive neighborhood (perhaps a historic district or university area) isolated within a sea of lower-priced areas.

- **Economic Interpretation:** The high price is driven by a unique local amenity or a highly restrictive local covenant, preventing the price effect from spreading.

Cont...

4. Low-High Outliers: A single, low-priced neighborhood adjacent to a thriving, high-priced area.

- **Economic Interpretation:** This is a transition zone or an area that has yet to gentrify.
- **Action:** Monitor this area for signs of rapid change; this is where pressure for redevelopment and displacement is likely to be highest.

Geary's C

- Geary's C is a statistic used to assess spatial autocorrelation, much like Moran's I, but it is more sensitive to local variations.
- Unlike Moran's I, which emphasizes overall similarity, Geary's C focuses on the degree of dissimilarity between neighboring values. For this reason, it is often described as a measure that captures negative spatial autocorrelation.

Cont...

$$C = \frac{(n-1)}{2W} * \frac{\sum_i \sum_j (Z_i - \bar{Z})(Z_j - \bar{Z})}{\sum_i (Z_i - \bar{Z})^2}$$

Geary's C Value	Interpretation (Pattern)
Close to 0	Positive Spatial Autocorrelation (Clustered): Similar values are near one another. High values cluster with high values; low values cluster with low values.
Close to 1	Random Distribution: There is no discernible pattern. The value of a feature is independent of its neighbors' values.
Close to 2	Negative Spatial Autocorrelation (Dispersed): Dissimilar values are near one another. High values are near low values (a checkerboard pattern).

Cont...

- The numerator essentially sums the squared differences between the values of neighboring pairs. The denominator is a normalization factor related to the total variance of the data.

Interpretation and Usage

- Like Moran's I, Geary's C is an inferential statistic and its value must be evaluated for statistical significance using a Z-score and p-value.
- Null Hypothesis (H_0): The attribute being analyzed is randomly distributed in space.

Interpreting Significance

- C close to 0 (and significant): The spatial pattern is clustered. You reject H_0 .
- C close to 2 (and significant): The spatial pattern is dispersed. You reject H_0 .
- C close to 1 (or not significant): The spatial pattern is random. You fail to reject H_0 .

Cont...

Feature	Global Moran's I	Local Moran's I	Geary's C
Type	Global statistic	Local statistic	Global statistic
Output	One value for entire area	Value for each location	One value for entire area
Focus	Overall similarity	Local clusters & outliers	Neighbor differences
Range	-1 to +1	Varies by location	0 to ~2
Random value	0	Not fixed	1

Example

Geary's C in Environmental Science (Soil Quality)

- Scenario: Mapping Soil Contamination
- An environmental consulting firm is studying a large industrial area to assess the spatial pattern of a heavy metal contaminant (like lead) in the soil. They are interested in seeing if the contamination levels change Suddenly or unexpectedly over short distances, which would suggest localized, point-source pollution.
- **Variable of Interest (xi):** Concentration of lead (ppm) measured at various sampling plots.
- **Spatial Units (n):** The individual sampling plots across the study area.
- **Spatial Weights (wij):** Usually based on **Inverse Distance**, where closer plots have a stronger influence on one another.

Application of Global Geary's C

- The firm calculates the **Global Geary's C** for the entire study area.

Calculated Value	Conclusion	Interpretation
C = 1.68	Significant Negative Autocorrelation (Z-score significantly low, $p < 0.05$)	This indicates a dispersed pattern. Plots with high lead concentrations tend to be right next to plots with low lead concentrations, and vice-versa.

- Implication for Cleanup:** This finding suggests that the pollution is highly localized and likely comes from many distinct point sources (e.g., small spills, leaky barrels, specific processing areas) rather than a single, widespread source (like acid rain or a general flood).
- The “inconsistent” nature means that cleanup efforts must be focused and targeted at small areas, as the pollution levels change rapidly over space.

Comparison with Moran's I in this Context

- If the firm had instead calculated Global Moran's I for this same data, the result would likely be a value close to zero (e.g., $I \approx -0.3$).
- A low Moran's I would suggest randomness.
- A high Geary's C (closer to 2) explicitly confirms dispersion or high dissimilarity between neighbors.
- Why the difference? In cases of dispersion (like the checkerboard pattern), Moran's I tends to be closer to zero and may suggest a random pattern, whereas Geary's C is specifically designed to detect this dissimilarity between adjacent units, making it sometimes a better choice for detecting localized differences or negative autocorrelation.

Geary's C vs. Moran's I

- While both measure spatial autocorrelation, they can sometimes yield slightly different results because they emphasize different things:

Feature	Moran's I (I)	Geary's C (C)
Comparison	Value vs. Global Mean	Value vs. Neighbor's Value
Sensitivity	Sensitive to extreme values (outliers) across the entire study area.	Sensitive to local differences between neighbors.
Range	Typically $[-1,1]$	Typically $[0,2]$
Clustered Pattern	High positive I	Low positive C

3.3. Spatial heterogeneity

- The term spatial heterogeneity refers to variation in relationships over space. In the most general case we might expect a different relationship to hold for every point in space.
- We introduce two principal mathematical quantities which serve as measures of association for spatially dependent data (which is Covariance function and Variogram).

Covariance function

- The **covariance function** describes how a spatial variable $Z(s)$ relates to itself at two different locations s_i and s_j :

$$\text{Cov}[Z(s_i); Z(s_j)] = E[Z(s_i); Z(s_j)] - E[Z(s_i)]E[Z(s_j)]$$

Purpose

- To understand the spatial continuity and dependence structure of a spatial random field.
- Assumptions (Second-Order Stationarity)
- Constant mean: $E(Z(s_i)) = \mu$,
- Covariance depends only on separation distance ($h = s_i - s_j$, NOT location s_i)
- That is, covariance depends on distance h , not on the specific locations.

3.4. Variogram , semi-variogram, nugget, range, and sill

- The variogram is the cornerstone of geostatistical analysis. It's a plot that shows how the spatial dependence or "dissimilarity" between data points changes as the distance between them increases.
- The variogram ($2\gamma(h)$) and semi-variogram ($\gamma(h)$) are essential tools for quantifying spatial correlation, particularly for geostatistical interpolation methods like Kriging .

Variogram

$$2\gamma(s,h) = E[(Z(s) - Z(s+h))^2]$$

- It represents the expected squared difference between values separated by distance h.

Semivariogram

$$\gamma(s,h) = \frac{E[(Z(s) - Z(s+h))^2]}{2}$$

- The semivariogram is simply half of the variogram and is the version most commonly used in practice.

Empirical Semivariogram

- To estimate the semivariogram from data, we compute the average squared differences of all point pairs separated by distance h

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(s_i) - Z(s_i+h)]^2$$

Cont...

Where:

- $\gamma(h)$: semivariance at lag distance h
- $N(h)$: number of data pairs separated by distance h
- $Z(s_i)$: observed value at location s_i

Stationarity Requirement

- For the variogram to depend only on distance and not on absolute location, the data must satisfy the intrinsic stationarity assumption:
- $\gamma(h) = \gamma(|h|)$
- This means that dissimilarity depends only on how far apart points are, not where they are in the study area.

Relationship to Covariance

- The variogram can also be expressed using covariance:
- $2\gamma(s_i, s_j) = [\text{Var}\{Z(s_i)\} + \text{Var}\{Z(s_j)\} - 2 \text{Cov}\{Z(s_i), Z(s_j)\}]$
- As distance increases, covariance decreases, causing $\gamma(h)$ to increase until it reaches the sill.

Components of a Variogram

- **Nugget Effect (C_0):** The non-zero value of the semivariogram at a separation distance of $h=0$. It represents unexplained variability at distances smaller than the minimum sampling distance, which may be due to measurement error or micro-scale variation.
- **Sill (C_0+C):** The constant value that the semivariogram approaches as the separation distance (h) increases. The sill typically represents the total variance (σ^2) of the data if the spatial process is stationary.
- **Range (α):** The separation distance (h) at which the semivariogram first reaches the sill (Menezes et al., 2005). At this distance and beyond, the data values are considered spatially uncorrelated.
- The semivariogram plot, which graphs $\gamma(h)$ against the separation distance ($|h|$), reveals key parameters of the spatial structure:

Cont...

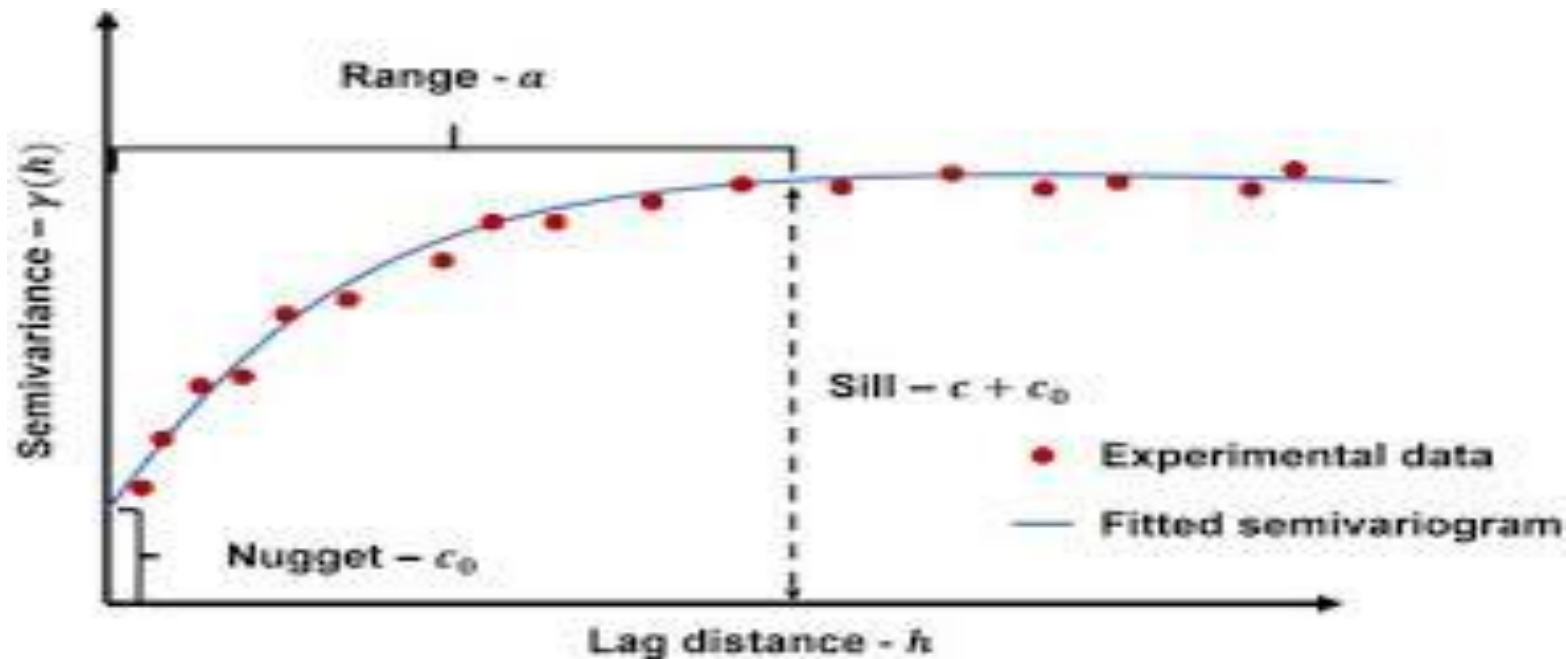


FIGURE 3.2: Typical semivariogram.

Interpretation and Use

- The variogram provides a measure of **spatial autocorrelation** (or spatial continuity):
- **Low Semivariance at Small Distances:** Points close together have small differences in value, indicating a **strong positive correlation** (they are similar). This is consistent with **Tobler's First Law of Geography**: "everything is related to everything else, but near things are more related than distant things."
- **Increasing Semivariance:** As distance increases (up to the range), the semivariance increases, meaning the correlation weakens and the points become more dissimilar.
- **Flat Semivariance (at the Sill):** Beyond the range, the values are essentially uncorrelated, and the variance remains constant, similar to what you'd expect from a random set of independent data.

3.5. Isotropic and Anisotropic Analysis

- Spatial dependence may vary with direction.
- **Isotropic:** The spatial variability, as characterized by the semivariogram, depends only on the distance between two locations (h), regardless of the direction. The structure of the spatial dependence is the same in all directions.
 - e.g., soil properties vary smoothly and uniformly in all directions.
- **Anisotropic:** The semivariogram depends on both the distance and the direction between two locations. The spatial continuity is different in different directions.
 - e.g., wind erosion or geological strata cause directional variation.
- **Geometric Anisotropy:** The range of the variogram changes with direction, but the sill remains constant.
- **Zonal Anisotropy:** The sill of the variogram changes with direction, while the range can remain the same
- Identifying anisotropy is crucial for accurate modeling and kriging.

3.6 Types of Variograms

- Experimental variogram: Computed from data.
- Theoretical variogram model: Fitted to the experimental variogram for prediction (e.g., spherical, exponential, Gaussian models).
- Example 1: Let's compute a simple empirical variogram using a tiny dataset.
- Data: Assume we have 5 points on a line:

Point	Position (s)	Value (z)
1	0	10
2	1	12
3	2	13
4	4	20
5	5	22

Cont...

Step 1 — Choose lag classes

- Let's use lag distance = 2 units:
- Lag 1: 0–2
- Lag 2: 2–4
- Lag 3: 4–6

Step 2 — Compute semivariance for each lag

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(s_i) - Z(s_j)]^2$$

Lag 1 (0–2 distance) pairs:

Pairs and squared differences:

- (1,2): $(10-12)^2=4$; (2,3): $(12-13)^2=1$; (3,4): distance = 2 $\rightarrow$ included $\rightarrow (13-20)^2=49$
- Total = $4 + 1 + 49 = 54$
Number of pairs = 3
- $\gamma(1) = \frac{54}{2*3} = 9$

Cont...

Lag 2 (2–4 distance) pairs

Pairs:

- (1,3): $(10-13)^2=9$
- (2,4): $(12-20)^2=64$
- (3,5): $(13-22)^2=81$
- Total = 154, $N = 3$
- $\gamma(2)=154/6\approx 25.67$

Lag 3 (4–6 distance)

Pairs:

- (1,4): $(10-20)^2=100$
- (1,5): $(10-22)^2=144$
- (2,5): $(12-22)^2=100$
- Total = 344, $N = 3$
- $\gamma(3)=344/6\approx 57.33$

Cont...

Lag distance	Semivariance
1	9
2	25.67
3	57.33

- This increases with distance showing spatial correlation decays over space.

2. How to Fit Variogram Models

- We fit a smooth theoretical model to the empirical points. The typical variogram parameters:
- Nugget (C_0): value at distance 0
- Sill ($C+C_0$): plateau value
- Range (a): distance where variogram flattens

From our empirical example:

- Nugget ≈ 0 (first semivariance small relative to others)
- Sill $\approx$ around 60

Cont...

- Range $\approx$ **around distance 3–4**

How to compute directional variograms

- Compute semivariance **only using pairs in a narrow direction band.**
- For example:
- $0^\circ \pm 22.5^\circ$ (east-west)
- $90^\circ \pm 22.5^\circ$ (north-south)
- Plot separate variograms for each direction.

Interpreting anisotropy

- If the 0° variogram reaches the sill at 20 m, but the 90° variogram reaches it at 50 m, then:
- The spatial correlation lasts longer north–south
- The phenomenon is elongated in that direction

Chapter 4

Spatial interpolation and prediction

4.1 Concept of Spatial Interpolation

- **Spatial interpolation** is the process of estimating the values of a variable at **unsampled locations** within a study area, based on the known values at a set of measured control points.

We need spatial interpolation to:

- To create **continuous surfaces** (e.g., rainfall, temperature, elevation).
- To fill gaps in spatial datasets.
- To support spatial decision-making for fields such as environmental science, agriculture, epidemiology, and resource management.

Basic Assumptions

- Spatial autocorrelation: nearby values tend to be more alike.
- A smooth or predictable pattern exists across space.

4.2. Exploring Different Interpolation Methods

- Spatial interpolation techniques can be broadly classified into **deterministic** and **geostatistical** methods.

Deterministic vs. Geostatistical:

- **Deterministic:** These methods are based solely on the degree of similarity or proximity. They use explicit mathematical formulas to determine the smoothness and shape of the resulting surface (e.g., IDW, Spline).
- **Geostatistical:** These methods, most notably Kriging, are based on statistical models that include spatial autocorrelation. They incorporate the spatial covariance structure of the data and provide a measure of the prediction uncertainty.

Deterministic Interpolation Methods

Inverse Distance Weighting (IDW)

- Inverse Distance Weighting (IDW) is a deterministic spatial interpolation method used to estimate values at unsampled locations based on nearby known values. It is built on the assumption that:
- Points closer to the prediction location have more influence than points farther away.
- This makes IDW one of the simplest and most widely used interpolation techniques.

Basic Principle of IDW

- **IDW** is a deterministic, exact interpolation technique that assumes the influence of a known point on an unknown point decreases with distance.
- The unknown value is a weighted average of the known sample values.
- The weight assigned to each known point is inversely proportional to its distance from the prediction location. Closer points receive higher weights, and farther points receive lower weights.
- The interpolated surface is smooth and continuous.

IDW Formula

- The value Z at an unmeasured location s_0 is calculated as a weighted average of the values Z_i at nearby measured locations s_i .
- The predicted value at an unknown location s_0 is computed as:

$$Z(s_0) = \frac{\sum_{i=1}^n Z_i d_i^{-p}}{\sum_{i=1}^n d_i^{-p}}$$

- where d_i is the distance between the unknown point (s_0) and the i^{th} measured point, and p is the power parameter (usually 2, for inverse distance squared).

Cont....

- **Pros:** Simple, fast, and easy to understand.
- **Cons:** Does not account for spatial structure (autocorrelation) and often results in a "bull's-eye" effect around sample points. Predictions never exceed the maximum or fall below the minimum sampled value.

The Power Parameter (p)

- The parameter p controls the degree to which distance affects weighting:

When $p=1$

- Gradual decay of influence.
- Produces smoother surfaces..

Cont...

When $p=2$ (most common)

- Influence decreases more sharply with distance.
- Nearby points dominate prediction.

When $p>2$

- Strong influence from very close points.
- Can lead to "bull's-eye" patterns.

When $p<1$

- Distant points exert more influence.
- Surface becomes overly smooth.

Steps in IDW Interpolation

- Identify the prediction location.
- Select a set of surrounding sample points (e.g., nearest 12 points or within a search radius).
- Calculate distances from each sample point to the prediction point.
- Apply the IDW formula with chosen p .
- Generate predicted value.
- Repeat for each grid cell to create a continuous surface.

Spline Interpolation

- Spline interpolation is a deterministic method that generates a smooth surface by fitting a series of mathematical functions (specifically, piecewise polynomials) through the known input points.
- The core idea is to create a surface that is both continuous and has the minimum possible curvature, like bending a thin, elastic sheet of metal or rubber to pass exactly through the measured data points.
- It is like bending a flexible sheet to fit the data.

Conceptual Background

- **Minimizing Curvature:** The function used is designed to minimize the overall "roughness" of the surface. This results in a smooth, continuous surface.
- **Exact Interpolator:** Like Inverse Distance Weighting (IDW), the interpolated surface **must pass exactly** through every single input data point. The predicted value at a sample location will equal the measured value.
- **The Analogy:** Historically, a "spline" was a flexible strip of wood or metal used in shipbuilding to draw smooth curves through a set of predefined points. Mathematical spline interpolation mimics this physical process.

Cont...

- Best Use: Spline is highly suitable for creating surfaces of phenomena that change gradually and smoothly, such as:
 - Elevation (Digital Elevation Models - DEMs)
 - Water table heights
 - Pollution concentrations over a large area
- Pros: Creates a very smooth, aesthetically pleasing surface.
- Cons: Can create large prediction errors or unrealistic peaks/valleys in areas where data is sparse, and it does not account for measurement error.

Geostatistical Prediction: Kriging

- Kriging is an advanced **geostatistical interpolation technique** used to predict values at unsampled locations based on the spatial autocorrelation of measured data.
- Weights depend on spatial configuration and **variogram**, not arbitrary distance powers.

It is widely considered the most powerful interpolation method because it provides:

- **Best Linear Unbiased Estimator (BLUE)**
- A measure of prediction **uncertainty (kriging variance)**
- Flexible adaptation to spatial patterns and trends

Core Principles

1. Modeling Spatial Structure: Kriging uses the **variogram** to quantify the spatial autocorrelation and model how the variance changes with distance.

2. Estimation: It calculates the unknown value $\hat{Z}(s_0)$ as a weighted sum of the known values $Z(s_i)$.

3. Optimization: The weights (λ_i) are chosen to ensure the estimate is the **Best Linear Unbiased Estimator (BLUE)**, meaning:

- **Unbiased:** The expected prediction error is zero. $E[\hat{Z}(s_0) - Z(s_0)] = 0$
- **Linear:** weighted linear combination of observed values.
- **Best:** The estimation variance (σ_k^2) is minimized.

Cont...

- Kriging provides a prediction at an unsampled location as a **weighted linear combination** of known sample data points. The key assumption is local mean is *constant but unknown*.

$$\hat{Z}(s_o) = \sum_{i=1}^n \lambda_i Z(s_i)$$

Where:

- $Z(s_i)$ = known data value at location s_i .
- λ_i = weights assigned to each sample.
- $\hat{Z}(s_o)$ = predicted value at unsampled location s_o .
- n is the number of observed data points used in the estimation.

Types of Kriging

- **Simple Kriging (SK):** Assumes the mean is known and constant over the entire study area. Rarely applicable in practice as the true mean is known.
- **Ordinary Kriging:** Assumes the mean (μ) of the data is **unknown but constant** across the study area. This is the most widely used form.
- **Universal Kriging:** Assumes the mean (μ) is **not constant** and varies according to a spatial trend (e.g., elevation decreases linearly from North to South).

Using Variograms for Kriging

- Kriging requires a **variogram**(describes how similar or different data values are as the distance between them increases.) **model**,
- Workflow from variogram $\rightarrow$ kriging

Cont...

1. Compute **empirical variogram**
2. Fit **theoretical model**
3. Compute the **covariance function** (variogram and covariance are linked)
4. Build the **kriging system**:

$$C\lambda = d$$

Where:

- C = covariance matrix between known points
 - d = covariance vector between known points and prediction location
 - λ = kriging weights
5. Solve for the kriging weights

Cont...

6. Predict value:

$$\hat{Z}(s_o) = \sum_{i=1}^n \lambda_i Z(s_i)$$

7. Compute kriging variance:

- $\sigma_k^2 = C(0) - \lambda^T d$
- This gives both the predicted value **and** its uncertainty.

4.3 Uncertainty Assessment in Spatial Interpolation

- Spatial interpolation involves estimating values at unsampled locations using observations from sampled points.
- Because these predictions are not exact, **uncertainty assessment** is essential to evaluate the reliability of the predicted spatial surfaces.
- A critical advantage of **Kriging** over deterministic methods (like IDW) is its ability to provide a measure of prediction uncertainty.

Why Assess Uncertainty?

- Prediction without uncertainty quantification is incomplete and potentially dangerous for decision-making. Uncertainty assessment is crucial and helps answer:

Cont...

- Where are my predictions most/least reliable?
- How might errors propagate into subsequent analyses?
- Should I collect more data, and if so, where?

Sources of Uncertainty

- **Measurement Error:** Errors in the sample data collection (e.g., instrument precision, recording mistakes).
- **Sampling Error:** The uncertainty introduced by the spatial design of the sample (e.g., a sparse or clustered sampling pattern).
- **Model Error:** The error introduced by the choice of interpolation method (e.g., an IDW surface will inherently be different from a Kriging surface).
- **Spatial Variation:** The natural, unpredictable heterogeneity of the variable being mapped.

Common Measures of Uncertainty

1. Kriging Variance (σ_k^2)

- **Definition:** The Kriging variance is the output of the Kriging equations, representing the **estimated error** or **uncertainty** at the predicted location.

Key Feature: Unlike the actual prediction error, the Kriging variance **does not depend on the data values**, only on:

- The **spatial configuration** of the sample points.
- The **distance** from the prediction point to the nearest samples.
- The parameters of the **variogram model** (Nugget, Sill, Range).

Use: Mapping the Kriging variance creates an **uncertainty surface**. Areas farther from sampled points or with high Nugget effects will have higher variance (greater uncertainty).

2. Cross-Validation

- **Definition:** A general technique used to assess the accuracy of **any** interpolation model.
- **Mechanism:** Each sample point is removed, and the model is used to predict the value at that location using the remaining points. The predicted value is then compared to the true observed value.

Metrics: Commonly used metrics derived from cross-validation include:

- **Mean Error (ME):** Should be close to zero (checks for bias).
- **Root Mean Squared Error (RMSE):** Measures the average magnitude of the error.

4.4 Predictive Modeling using Spatial Data

- Predictive modeling with spatial data involves creating models that forecast outcomes by explicitly incorporating the **geographic location** and **spatial relationships** of the data points.
- The goal is to produce accurate predictions across a continuous geographic surface, often overcoming limitations of traditional non-spatial models which can be biased when data exhibits spatial patterns. Two key challenges addressed are:
 - **Spatial autocorrelation** – nearby observations are similar; corrected using spatial regression models (spatial lag model(SLM), spatial error model(SEM)).
 - **Spatial heterogeneity** – relationships vary across space; addressed with Geographically Weighted Regression (GWR).

Main Spatial Predictive Techniques

1. Co-Kriging

- Uses multiple spatially correlated variables to improve predictions. A primary variable (sparse) is predicted using a secondary, more densely sampled variable.
- **Principle:** An extension of Kriging that uses **multiple variables** to improve prediction accuracy.
- **Mechanism:** It requires that the primary variable of interest (Z_1) is spatially correlated with a secondary variable (Z_2) that is **easier or cheaper to sample** more extensively. Co-Kriging models the spatial autocorrelation of Z_1 and Z_2 , as well as the **cross-correlation** between them.

Cont...

- **Example:** Predicting soil lead concentration (primary variable, sparsely sampled) using soil cadmium concentration (secondary variable, densely sampled) and their known cross-correlation.

2. Spatial Regression

- These are extensions of traditional regression used to model relationships (not just interpolate values) while accounting for spatial dependence.

Spatial Lag Model (SLM) / Spatial Autoregressive Model (SAR):

- **Focus:** Models spatial autocorrelation in the **dependent variable** (Y).
- **Concept:** The outcome at location i is directly influenced by the outcome at neighboring locations.
- **Use Case:** Modeling the spread of an infectious disease or housing prices, where neighbors directly affect each other.

Cont..

- **General Form:** $Y = \rho WY + X\beta + \epsilon$

Spatial Error Model (SEM):

- **Focus:** Models spatial autocorrelation in the **error term** (ϵ).
- **Concept:** The model's unobserved factors (errors) are spatially clustered, indicating missing spatially clustered covariates.
- **Use Case:** Predicting crop yield using known variables, but unmeasured factors like local soil drainage (which are spatially clustered) affect the error.
- **General Form:** $Y = X\beta + \lambda W\epsilon + u$

3. Geographically Weighted Regression (GWR)

- GWR is a technique for addressing **Spatial Heterogeneity** by fitting a separate, local regression equation for every data point in the dataset.
- **Mechanism:** GWR uses a **spatial kernel function** to assign a weight to every other observation for that local regression. Observations closer to the target point receive a higher weight and thus have more influence on the local coefficient estimates.
- **Output:** Instead of a single global coefficient for each predictor, GWR produces a set of coefficients (and an intercept) for *each location*. These can then be mapped to visualize how the relationship changes across space.
- **Advantage:** It acts as a "spatial microscope," revealing local variations that a single global model would mask.

4. Geostatistical & Hybrid Models

Used for mapping continuous surfaces.

- **Universal Kriging:** includes trends (e.g., elevation).
- **Regression-Kriging:** combines regression for large-scale trends with Kriging of residuals for local variation. This is a powerful, two-step hybrid approach:
 - ✓ Fit a standard regression model (e.g., OLS) using the available covariates (X) to model the **large-scale trend**.
 - ✓ Use **Ordinary Kriging** on the **residuals** (the error term) from the regression model to model the **short-range spatial correlation**.
 - ✓ The final prediction is the sum of the predicted regression trend and the interpolated residual surface.

5. Spatial Machine Learning

- Applies algorithms like Random Forest or Gradient Boosting using spatial features (coordinates, distances) and spatial cross-validation to avoid overfitting to nearby samples.
- Traditional ML algorithms (Random Forest, Gradient Boosting) can be adapted for spatial prediction by:

Feature Engineering: Explicitly including spatial coordinates (X, Y) or other derived spatial metrics (e.g., distance to nearest feature, neighborhood density) as input features.

Spatial Cross-Validation: Using spatial blocking or buffer-based methods for training/testing splits to ensure that the model is tested on locations geographically separate from the training data. This provides a more honest assessment of the model's true predictive power in new areas.

Chapter 5

Spatial Autoregressive Models

5.1 Concepts of spatial autoregressive models

- Spatial autoregressive (SAR) models are a type of spatial regression model used to analyze data where the value of a variable at one location is influenced by the values of the same variable at nearby locations. This phenomenon is known as **spatial dependence** or **spatial autocorrelation**.
- Traditional regression models assume that observations are independent of each other. However, in spatial data (e.g., crime rates in different neighborhoods), this assumption is often violated. Ignoring spatial dependence can lead to biased and inefficient model estimates. SAR models address this by incorporating the spatial relationships directly into the model's structure.

5.1.1 Spatial Dependence

- Spatial dependence (or spatial autocorrelation) means that the value of a variable at one location depends on values at nearby locations.
- **Positive spatial dependence:** similar values cluster together
- **Negative spatial dependence:** dissimilar values are neighbors

Ignoring spatial dependence can lead to:

- Biased estimates
- Underestimated standard errors
- Incorrect inferences

5.1.2 Spatial Neighbors and Spatial Weights Matrices

- The concept of "spatial neighbors" is central to spatial autoregressive models. Neighbors are defined not just by physical proximity, but by a specified relationship.

Spatial Neighbors

Criteria for defining neighbors:

- **Contiguity-based:** Shared borders/edges (Queen vs. Rook criteria)
- **Distance-based:** Within specified distance threshold
- **K-nearest neighbors:** Fixed number of closest neighbors
- **Network-based:** Along transportation or infrastructure networks

Spatial Weights Matrix (W)

- **Definition:** An $n \times n$ matrix (where n = number of spatial units) describing spatial relationships

Elements (w_{ij}):

- $w_{ij} > 0$ if locations i and j are neighbors
- $w_{ij} = 0$ otherwise (including diagonal elements $w_{ii} = 0$)

Common normalizations:

- Row-standardization: Each row sums to 1 $\rightarrow$ easier interpretation
- Binary: 1 for neighbors, 0 otherwise
- Inverse distance: Weight decreases with distance

5.2. Types of Spatial Autoregressive Models

- There are several types of spatial models depending on where spatial dependence appears in the model.

5.2.1 Spatial Autoregressive Model (SAR or Lag Model)

- **Spatial Autoregressive (SAR) Model:** Also known as the spatial lag model, this model accounts for spatial dependence in the **dependent variable**.
- It assumes that the value of the outcome at a location is directly influenced by the values of the outcome at neighboring locations.
- For example, a high crime rate in one neighborhood might directly "spill over" and cause a higher crime rate in an adjacent neighborhood.
- The model includes a spatial lag term ($\rho W y$), where ρ is the spatial autoregressive coefficient.

Cont....

Model:

- $Y = \rho W_y + X\beta + \varepsilon$

Where:

- Y: dependent variable
- W_y : spatial lag of the dependent variable
- ρ : spatial autoregressive coefficient
- X: matrix of explanatory variables
- β : coefficients
- ε : error term

Cont...

Interpretation:

- ρ captures how much the dependent variable at a location is influenced by neighboring values.
- **Direct effects:** influence of predictors on a location
- **Indirect (spillover) effects:** influence of predictors on neighboring locations

5.2.2 Spatial Error Model (SEM)

- **Spatial Error Model (SEM):** This model accounts for spatial dependence in the **error terms**.
- It assumes that the spatial autocorrelation is a nuisance, likely caused by unobserved variables or measurement errors that are spatially clustered.

Cont...

- For example, if a model for home prices doesn't include a variable for "school quality," the uncaptured effect of good schools might show up as spatially clustered errors.
- The model includes a spatially lagged error term ($\lambda W\epsilon$), where λ is the spatial error coefficient.

Model:

- $Y = X\beta + u$, with $u = \lambda Wu + \epsilon$

Where:

- λ : Spatial error coefficient.
- **Wu** : The spatial lag of the errors (the "average error" of the neighbors).
- u : spatially autocorrelated error term
- ϵ : The final, truly random noise that is left over.

Interpretation:

- SEM is used when **unobserved spatial factors** influence the dependent variable.
- Doesn't model spillover effects in Y, only in the errors.

5.2.3 Comparison: SAR vs. SEM

Feature	SAR (Lag Model)	SEM (Error Model)
Spatial Dependence	In the dependent variable	In the error term
Interpretation	Direct and indirect effects (spillover)	Captures omitted spatial structure
Use When	Outcome is influenced by neighbors	Spatial dependence is in unobserved factors
Spatial Lag of Y	Included	Not included

5.3. Model Estimation and Interpretation

5.3.1 Estimation Methods

- Estimating spatial autoregressive models is more complex than standard Ordinary Least Squares (OLS) regression because of the feedback loops created by spatial dependence.
- **Estimation:** OLS is not appropriate for SAR models because the spatial lag term ($\rho W y$) makes the dependent variable correlated with the error term, leading to biased and inconsistent estimates.
- Instead, methods like Maximum Likelihood Estimation (MLE) or Generalized Method of Moments (GMM) are used.

5.3.2 Interpretation of Parameters

SAR Model: The coefficients of the independent variables cannot be interpreted directly as in OLS.

- A change in an independent variable at one location affects the dependent variable at that location, which then "spills over" and affects the dependent variable in neighboring locations, which in turn affects their neighbors, and so on.
- Therefore, it's essential to calculate direct effects (the impact on the same location) and indirect effects (the spillover impact on neighboring locations).

SEM Model: The coefficients of the independent variables can be interpreted more like OLS, as the model assumes the spatial dependence is confined to the error term.

The spatial error coefficient (λ) indicates the strength of the spatial autocorrelation in the residuals. A significant λ suggests that the model is missing important, spatially-correlated variables or that there is a spatial process in the error term that needs to be accounted for.

Cont...

- ρ (SAR): Positive value indicates **positive spatial spillover** (e.g., crime in one area influences neighboring areas).
- λ (SEM): Captures the **magnitude of spatial autocorrelation in errors**.
- β : Regression coefficients; in SAR, interpret with care due to feedback effects.

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5.3.3 Impacts in SAR Models

- Unlike OLS, SAR model coefficients **cannot be directly interpreted** as marginal effects due to the spatial feedback loop.
- You must calculate:
- **Direct effects:** Impact of X on its own location
- **Indirect effects:** Impact of X on neighboring locations
- **Total effects:** Sum of direct and indirect effects
- These are derived from the matrix:

$$(I - \rho W)^{-1} X \beta$$

Chapter 6

Applications of Spatial Statistics

Introduction

- Spatial statistics has diverse applications across many fields by leveraging the power of location and spatial relationships to gain new insights.
- The ability to analyze data in a geographical context helps researchers and practitioners understand patterns, predict outcomes, and make more informed decisions.

1. Environmental Science

- Spatial statistics is fundamental to environmental science. It is used to model and predict the distribution of pollutants, assess environmental risks, and manage natural habitats.
- For example, geostatistical methods like **kriging** are used to create continuous pollution maps from sparse monitoring data, helping to identify "hot spots" that need intervention.
- Similarly, it's used to track the spread of invasive species, analyze deforestation patterns, and model the impacts of climate change on ecosystems.

2. Agriculture and Forestry

- In agriculture and forestry, spatial statistics underpins the concept of **precision agriculture**. It helps farmers optimize resource use and maximize yields by analyzing spatial variation in soil properties, crop health, and nutrient levels across a field.
- **Yield prediction:** Spatial models can integrate soil data, weather patterns, and satellite imagery (e.g., NDVI) to predict crop yields and identify areas of stress.
- **Pest and disease management:** It's used to map the spread of pests and diseases, allowing for targeted and efficient application of pesticides or other control measures.
- **Forest management:** In forestry, it helps in inventorying tree species, assessing forest health, and planning sustainable logging operations.

3. Demography

- Demography, the study of human populations, is inherently spatial. Spatial statistics is used to analyze population distribution, migration patterns, and the spatial clustering of demographic variables.
- **Population change:** Geographically Weighted Regression (GWR) can be used to model how factors like income or education relate to population growth, revealing that these relationships can vary from one location to another.
- **Urban planning:** It helps urban planners understand population density, analyze access to public services, and plan for future growth and infrastructure needs.

4. Medicine and Public Health

- Spatial statistics, particularly in the field of **spatial epidemiology**, is crucial for understanding disease and health-related patterns.
- **Disease mapping:** The classic example is John Snow's 1854 cholera map, which used spatial data to identify the source of the outbreak. Modern applications include mapping disease incidence to identify clusters and predict future outbreaks.
- **Health resource allocation:** It helps public health officials determine the optimal locations for new clinics, hospitals, or vaccination centers to maximize accessibility and serve vulnerable populations.
- **Risk factor analysis:** Spatial models are used to investigate the relationship between environmental risk factors (e.g., air pollution) and health outcomes, helping to guide public health policy.

5. Mining and Petroleum Engineering

- Spatial statistics, often called **geostatistics** in this context, originated in the mining industry.
- **Resource estimation:** It is used to estimate the quantity and quality of ore reserves or oil and gas deposits by interpolating values from exploratory drill hole data.
- **Drilling optimization:** Variograms and kriging are used to model the spatial continuity of geological properties, which helps engineers decide where to place new exploratory or production wells to maximize resource extraction.
- **Risk assessment:** Spatial models help to assess geological risks, such as unstable slopes or sinkholes, to improve safety in mining operations.

6. Natural Resources

- Spatial statistics is essential for the sustainable management of a wide array of natural resources, including water, land, and wildlife.
- **Water management:** It's used to model groundwater levels, predict flood risks by analyzing elevation and rainfall data, and identify sources of water pollution.
- **Wildlife conservation:** Researchers use spatial point pattern analysis to study animal territories, migration routes, and habitat fragmentation. This information is vital for conservation planning and establishing protected areas.
- **Land use:** Spatial analysis helps to map and monitor land use changes, assess the impact of human activities on the environment, and plan for sustainable land development.

7. Geography

- As the foundational discipline for spatial statistics, geography utilizes these techniques to analyze and interpret all types of geographical phenomena.
- **Pattern analysis:** Geographers use spatial statistics to identify and explain patterns in everything from human migration to the distribution of plant species.
- **Spatial modeling:** It allows for the creation of predictive models that simulate geographical processes, such as urban spread of forest fires, helping to inform policy and planning.
- **GIS and mapping:** Spatial statistics provides the analytical engine behind Geographic Information Systems (GIS), enabling complex queries and sophisticated analyses that go far beyond simple visualization.